Worksheet 2.1 — Adding & Multiplying Polynomials

Grade 11 Functions, University (MCR3U) · Unit 2: Equivalent Algebraic Expressions

The Big Ideas

Two expressions are **equivalent** when they have the same value for every x — so they draw the *same graph*. Rewriting by collecting like terms and expanding products is how we turn one form into another without changing its value. Every rewrite below is tested: by substituting numbers, and by comparing graphs.

- **Like terms** (same variable part) combine; a minus sign before a bracket flips *every* sign inside.
- **Distribute:** every term of one factor meets every term of the other. The **area model** keeps score; FOIL is its shortcut for two binomials.
- **Special products:** $(a \pm b)^2 = a^2 \pm 2ab + b^2$ and $(a+b)(a-b) = a^2 - b^2$.
- **Degrees:** the degree of a product is the *sum* of the degrees; the degree of a sum is at most the larger one.
- **Test it:** one mismatch proves two expressions are *not* equivalent; a match at every tested value (or a single overlapping graph) supports — and algebra proves — equivalence.

Concept Essentials

Equivalent means the same graph

Two expressions are equivalent if they give the same output for every x. Rewriting (collecting like terms, expanding) must never change the value: **test** a rewrite by substituting numbers — one mismatch proves it wrong.

Expanding

Every term in one factor multiplies every term in the other. An **area model** lays the partial products in a grid; **like terms** (same colour) are then collected. Degrees *add* when you multiply.

Special products

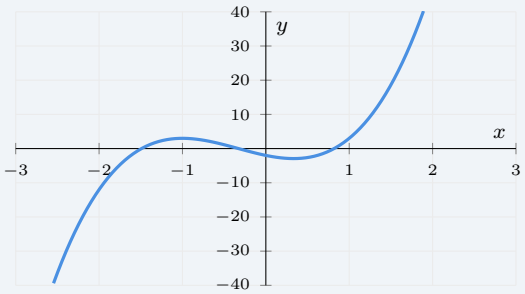
$(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$, $(a+b)(a-b) = a^2 - b^2$. They are shortcuts you will use *backwards* when factoring in 2.2.

Worked Examples

Example 1 — Adding polynomials

Simplify $(4x^3 - 2x + 7) + (x^3 + 5x^2 - 3x - 9)$.

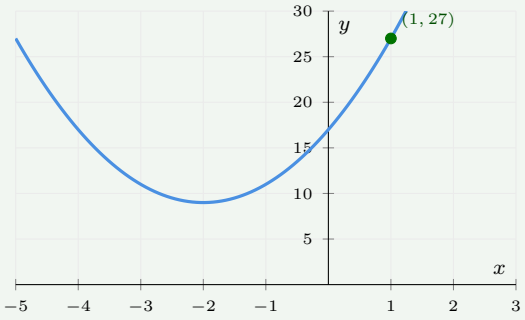
Solution. **Rule:** Adding keeps every sign; collect terms with the same power.
Group like terms by degree: $(4x^3 + x^3) + 5x^2 + (-2x - 3x) + (7 - 9)$.
$= 5x^3 + 5x^2 - 5x - 2$.
Check: at $x = 1$: $(4 - 2 + 7) + (1 + 5 - 3 - 9) = 9 - 6 = 3$, and the answer $5 + 5 - 5 - 2 = 3$ ✓



Example 2 — Distribute, then collect

Simplify $3(2x - 1) - 4(x - 5) + 2x(x + 3)$.

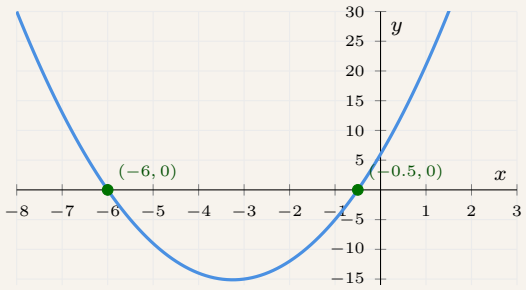
Solution. **Rule:** Distribute each outside factor, keeping its sign.
Distribute: $6x - 3 - 4x + 20 + 2x^2 + 6x$.
Note the -4 multiplies *both* terms: $-4 \cdot (-5) = +20$.
Collect: $2x^2 + 8x + 17$.
Check: at $x = 1$: $3(1) - 4(-4) + 2(4) = 3 + 16 + 8 = 27$, and the collected form gives $2 + 8 + 17 = 27$ ✓



Example 3 — Area model

Expand $(2x+1)(x+6)$.

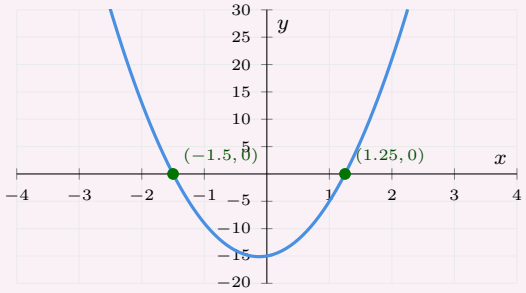
Solution. **Rule:** Each term of the first bracket meets each term of the second.
The boxes are $2x \cdot x = 2x^2$, $2x \cdot 6 = 12x$, $1 \cdot x = x$, $1 \cdot 6 = 6$.
The two x-boxes are like terms: $12x + x = 13x$.
$= 2x^2 + 13x + 6$.
Check: at $x = 1$: $(3)(7) = 21$, and $2 + 13 + 6 = 21$ ✓



Example 4 — FOIL with a negative

Expand $(4x-5)(2x+3)$.

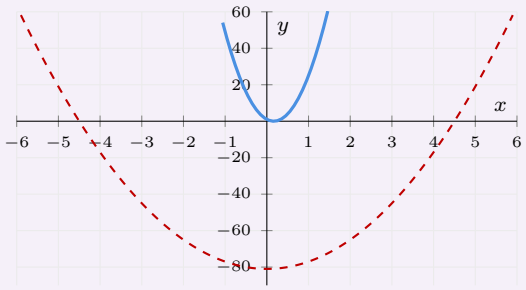
Solution. **Rule:** FOIL gives the four products; keep the signs.
First $8x^2$; Outer $12x$; Inner $-10x$; Last -15.
Middle: $12x - 10x = 2x$.
$= 8x^2 + 2x - 15$.
Check: at $x = 1$: $(-1)(5) = -5$, and $8 + 2 - 15 = -5$ ✓



Example 5 — Special products

Expand (a) $(6x-1)^2$ and (b) $(2x+9)(2x-9)$.

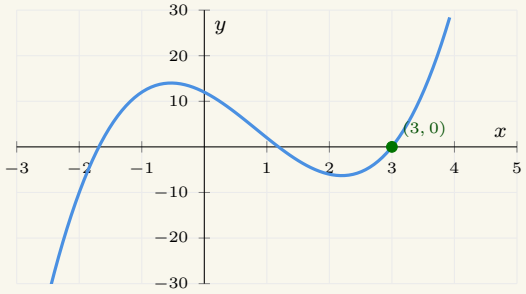
Solution. **Patterns:** $(a-b)^2 = a^2 - 2ab + b^2$ and $(a+b)(a-b) = a^2 - b^2$.
(a) $(6x)^2 - 2(6x)(1) + 1^2 = 36x^2 - 12x + 1$.
(b) The middle terms $\pm 18x$ cancel: $(2x)^2 - 9^2 = 4x^2 - 81$.
Check: at $x = 1$: (a) $25 = 36 - 12 + 1$; (b) $(11)(-7) = -77 = 4 - 81$ ✓



Example 6 — Trinomial times binomial

Expand $(2x^2 + x - 4)(x - 3)$.

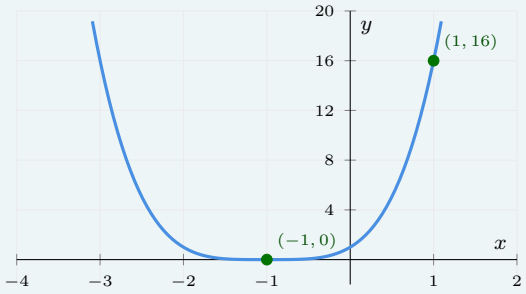
Solution. **Plan:** Three terms times two terms make six products.
Six partial products: $2x^3 - 6x^2 + x^2 - 3x - 4x + 12$.
Collect: $2x^3 - 5x^2 - 7x + 12$.
Check the degree $(2 + 1 = 3)$ and the constant term $(-4 \cdot -3 = 12)$.
Check: at $x = 1$: $(-1)(-2) = 2$, and the collected form gives 2 too ✓



Example 7 — A power of a binomial

Expand $(x+1)^4$.

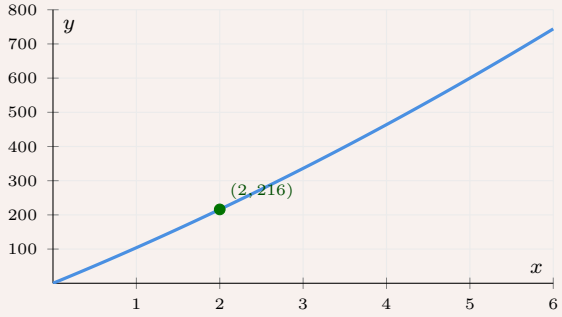
Solution. **Method:** Use Pascal's row for the exponent 4.
The coefficients are Pascal's row $1, 4, 6, 4, 1$ and every power of 1 is 1.
$= x^4 + 4x^3 + 6x^2 + 4x + 1$.
Check at $x = 1$: $2^4 = 16$ and $1 + 4 + 6 + 4 + 1 = 16$.



Example 8 — Application: a picture frame

A photo is 30 cm by 20 cm. A frame of width x goes all the way around it. Write the area of the frame alone.

Solution. **Model:** The outer rectangle gains x on each side, so $2x$ per dimension.
Outer rectangle: $(30 + 2x)(20 + 2x)$. Subtract the photo, 600.
$(30 + 2x)(20 + 2x) - 600 = 600 + 100x + 4x^2 - 600 = 4x^2 + 100x$.
For a 2 cm frame: $4(4) + 100(2) = 216$ cm^2.

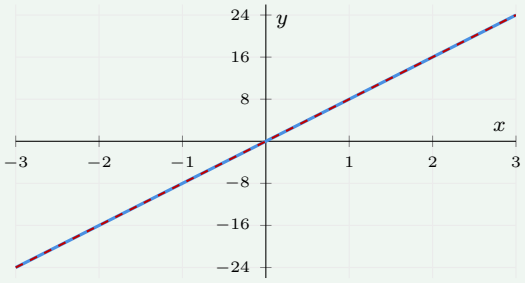


Example 9 — Are they equivalent?

Test whether $(x+2)^2-(x-2)^2$ and $8x$ are equivalent.

x	-2	-1	0	1	2
$(x+2)^2-(x-2)^2$	-16	-8	0	8	16
$8x$	-16	-8	0	8	16
same?	✓	✓	✓	✓	✓

Solution. The values match at all five points. To *prove* it, expand: $(x^2+4x+4)-(x^2-4x+4)=8x$ — the x^2 and constants cancel.



Both expressions draw the same line, so the dashed graph lies exactly on the solid one.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Simplify $(7x^2 - 3x + 4) - (3x^2 - x + 12)$. Then test your answer at $x = 2$.

Question 2

Expand $-3x(5x^2 - 2x - 1)$.

Question 3

Expand and simplify $(5x - 3)(2x + 7)$ using an area model.

Question 4

Expand (a) $(7x + 4)^2$ (b) $(5x + 6)(5x - 6)$. Explain why (b) has no x-term.

Question 5

Expand $(3x^2 - 2x - 6)(x - 4)$.

Question 6

Expand $(x - 4)^3$ using $(x - 4)^3 = (x - 4)(x - 4)^2$. Check your answer at $x = 5$.

Question 7

Simplify $(x + 9)^2 - (x - 3)^2$. Which special-product pattern gives a quicker route?

Question 8

Expand and simplify $(x + 2)(x - 3)(x + 4)$.

Question 9

Expand $(2x-3)(x^2-2x-8)$ and state the degree and the constant term *before* you multiply. Do they match your result?

Question 10

Complete the table to decide whether $(x-3)^2$ is equivalent to (a) x^2-9 or (b) x^2-6x+9. Then expand to confirm.

x	-1	0	1	2	3
$(x-3)^2$					
x^2-9					
x^2-6x+9					
same?					

Question 11

A student writes $(2x+1)^2-(x-3)(x+3)=3x^2+4x-8$. Find the error and give the correct simplified answer.

Question 12

A 24 cm by 18 cm sheet has squares of side x cut from each corner and the sides folded to make an open box. Write the volume as an expanded polynomial and find V when $x = 3$.

Question 13

Explain in a sentence or two why the degree of $(x^3 + x) + (-x^3 + 2x^2)$ is smaller than the degree of either bracket. Give your own example of a sum of two cubics that is a quadratic.

Question 14 — Challenge

Show that $n^2 + (n+1)^2$ is always odd, for any integer n. Then show that $(n+1)^2 - n^2$ equals the sum of the two numbers n and $n+1$.

Question Recap & Answer Key

Question Recap

1. Simplify $(7x^2 - 3x + 4) - (3x^2 - x + 12)$. Then test your answer at $x = 2$.
2. Expand $-3x(5x^2 - 2x - 1)$.
3. Expand and simplify $(5x - 3)(2x + 7)$ using an area model.
4. Expand (a) $(7x+4)^2$ (b) $(5x+6)(5x-6)$. Explain why (b) has no x-term.
5. Expand $(3x^2 - 2x - 6)(x - 4)$.
6. Expand $(x - 4)^3$ using $(x - 4)^3 = (x - 4)(x - 4)^2$. Check your answer at $x = 5$.
7. Simplify $(x + 9)^2 - (x - 3)^2$. Which special-product pattern gives a quicker route?
8. Expand and simplify $(x + 2)(x - 3)(x + 4)$.
9. Expand $(2x - 3)(x^2 - 2x - 8)$ and state the degree and the constant term *before* you multiply. Do they match your result?
10. Complete the table to decide whether $(x - 3)^2$ is equivalent to (a) $x^2 - 9$ or (b) $x^2 - 6x + 9$. Then expand to confirm.

x	-1	0	1	2	3
$(x-3)^2$					
x^2-9					
x^2-6x+9					
same?					

11. A student writes $(2x + 1)^2 - (x - 3)(x + 3) = 3x^2 + 4x - 8$. Find the error and give the correct simplified answer.
12. A 24 cm by 18 cm sheet has squares of side x cut from each corner and the sides folded to make an open box. Write the volume as an expanded polynomial and find V when $x = 3$.
13. Explain in a sentence or two why the degree of $(x^3 + x) + (-x^3 + 2x^2)$ is smaller than the degree of either bracket. Give your own example of a sum of two cubics that is a quadratic.
14. Show that $n^2+(n+1)^2$ is always odd, for any integer n. Then show that $(n + 1)^2 - n^2$ equals the sum of the two numbers n and $n + 1$.

✓ Answer Key

1. $4x^2 - 2x - 8$. At $x = 2$: original $(28 - 6 + 4) - (12 - 2 + 12) = 26 - 22 = 4$; answer $16 - 4 - 8 = 4$.
2. $-15x^3 + 6x^2 + 3x$.
3. $10x^2 + 29x - 21$ (boxes $10x^2$, $35x$, $-6x$, -21).
4. (a) $49x^2+56x+16$ (b) $25x^2-36$. The two middle terms $\pm 30x$ are opposites and cancel.
5. $3x^3 - 14x^2 + 2x + 24$.
6. $x^3 - 12x^2 + 48x - 64$. At $x = 5$: $(1)^3 = 1$ and $125 - 300 + 240 - 64 = 1$.
7. $24x + 72$. Difference of squares with $a = x + 9$, $b = x - 3$: $(a - b)(a + b) = 12(2x + 6) = 24x + 72$.
8. $x^3 + 3x^2 - 10x - 24$.
9. Degree $1 + 2 = 3$, constant term $(-3)(-8) = 24$. Result $2x^3 - 7x^2 - 10x + 24$ ✓.
10. Row values: $(x - 3)^2$: $16, 9, 4, 1, 0$; $x^2 - 9$: $-8, -9, -8, -5, 0$; $x^2 - 6x + 9$: $16, 9, 4, 1, 0$. So (b) is equivalent, (a) is not. Expanding $(x-3)(x-3)$ gives $x^2 - 6x + 9$.
11. $(2x+1)^2 = 4x^2+4x+1$ and $(x-3)(x+3) = x^2-9$, so the subtraction gives $4x^2 + 4x + 1 - x^2 + 9$. The student wrote -8 instead of $+10$ (forgot the minus flips -9 to $+9$). Correct: $3x^2 + 4x + 10$.
12. $V = x(30 - 2x)(20 - 2x) = 4x^3 - 84x^2 + 432x$. At $x = 3$: $3 \cdot 18 \cdot 12 = 648$ cm^3; check $108 - 756 + 1296 = 648$.
13. The x^3 terms cancel $(x^3 - x^3)$, leaving $2x^2 + x$. A sum can have lower degree than its parts. Example: $(x^3 + 4x) + (-x^3 + x^2) = x^2 + 4x$.
14. $n^2 + (n + 1)^2 = 2n^2 + 2n + 1 = 2(n^2 + n) + 1$, which is $2\times$ an integer plus 1: odd. And $(n + 1)^2 - n^2 = 2n + 1 = n + (n + 1)$.